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Received: 30 January 2026

Accepted: 27 May 2026

Published online: 30 May 2026

Cite this article as: Shi Y., Yin J., Ning S. *et al.* DeepCas12a: a hybrid deep learning framework for accurate AsCas12a efficiency prediction from sequence and epigenetic information. *BMC Genomics* (2026). <https://doi.org/10.1186/s12864-026-13003-3>

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DeepCas12a: A hybrid deep learning framework for accurate AsCas12a efficiency prediction from sequence and epigenetic information

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Abstract

CRISPR-Cas12a (Cpf1) offers distinct advantages for genome editing due to its flexible, T-rich PAM recognition. However, variable cleavage efficiency—modulated by sequence context and epigenetic features—remains a challenge, with existing tools facing challenges in modeling the high-order interactions between multimodal features. Here, we present DeepCas12a, a hybrid deep learning framework integrating Convolutional Neural Networks (CNNs) and a Vision Transformer (ViT) encoder to capture both local sequence motifs and long-range dependencies. The model fuses DNA sequence data with epigenetic profiles (DNA methylation and chromatin accessibility) in an end-to-end architecture. Benchmarked on an independent test set, DeepCas12a outperformed state-of-the-art predictors, achieving an Average Precision of 0.783, an AUC of 0.868, and a Spearman correlation of 0.630. Furthermore, interpretability analysis via saliency maps confirms the model captures biologically relevant features, including PAM specificity and seed region sensitivity, facilitating rational guide RNA design.

1. Introduction

Since its initial application in 2012[1], CRISPR genome editing has demonstrated transformative potential across agriculture, medicine, and basic research[2-4]. Among these, the type II CRISPR-Cas9 system has emerged as the dominant tool due to its high editing efficiency. However, limitations such as the restricted PAM requirements of commonly used Cas9 nucleases (e.g., NGG for SpCas9) and off-target effects limit their utility in complex genomic contexts[5]. Consequently, the type V CRISPR-Cas12a system has gained traction as an alternative, offering distinct mechanistic advantages that address the core limitations of Cas9[6].

Unlike Cas9, many commonly used Cas12a orthologs (such as AsCas12a) typically recognize a T-rich PAM (TTTV), significantly expanding the accessible genomic target space[7-9]. Furthermore, its reliance on a single crRNA simplifies construct design.

Despite these advantages, Cas12a orthologs display significant heterogeneity in cleavage efficiency, with variation exceeding an order of magnitude across different guide RNAs[10]. While current efficiency prediction tools, such as DeepCpf1, achieved state-of-the-art performance upon their release, their generalization capability is often limited when evaluated on cross-batch or out-of-distribution independent datasets, highlighting the need for novel architectures with stronger representational power[11]. [This performance gap stems from the mechanistic complexity of AsCas12a, where efficiency is modulated by crRNA- and spacer-related sequence features \[12\], local sequence context, target-site accessibility \[11\], and PAM-recognition mechanisms \[13\].](#) Furthermore, while robust predictive models exist for Cas9, the profound biological differences between the two nucleases—specifically in PAM recognition mechanisms, RNA structures, and cleavage patterns—prevent the direct application of existing Cas9 models to Cas12a. Therefore, dedicated AsCas12a prediction frameworks are required.

To address these challenges, we propose DeepCas12a, a deep learning framework utilizing a dual-stream hybrid architecture (Figure 1). This model integrates local feature extraction via Convolutional Neural Networks (CNNs) and global sequence modeling via a Transformer-based attention mechanism. Specifically, parallel convolutional layers capture motif preferences across the input sequence, while the Transformer encoder models long-range dependencies among sequence and epigenetic features. Benchmarking results demonstrate that DeepCas12a significantly outperforms existing tools in prediction accuracy on independent test sets.

Furthermore, we provide interpretable saliency maps to elucidate the mechanistic basis of AsCas12a activity. This framework offers practical guidance for therapeutic gene editing design—exemplified by targets for sickle cell disease—thereby reducing experimental screening costs.

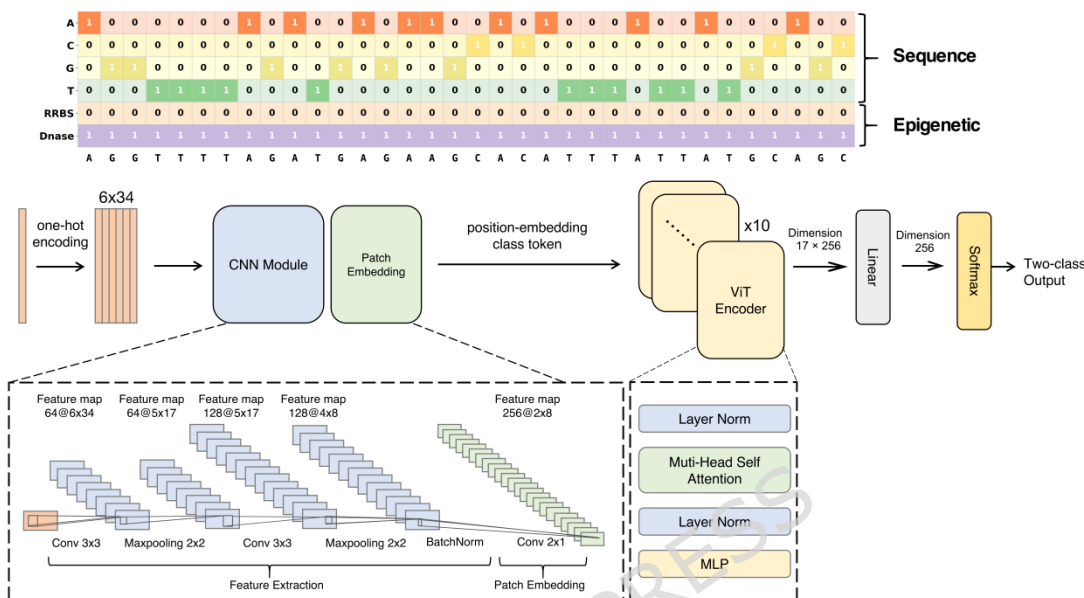


Figure 1. Architecture of the DeepCas12a Model Integrating Convolutional and Vision Transformer Modules for Predicting CRISPR-AsCas12a on-target cleavage efficiency.

2. Material and Methods

2.1 Data Preparation

CRISPR-AsCas12a on-target cleavage efficiency data were retrieved from Kim et al. (2018)[11]. This dataset (HT1) quantifies AsCas12a editing efficiency across target sites in HEK293T cells via high-throughput sequencing. The complete dataset (N=16,292) was randomly partitioned into a training set (n=15,000) and an independent test set (n=1,292).

In addition to HT1, two additional AsCas12a activity datasets from Kim et al. (2018) were incorporated as supplementary training data: the HEK-lenti dataset (n = 148) and the HEK-plasmid dataset (n = 55)[11]. Together, these datasets provided 203 additional AsCas12a activity records measured in HEK293T-based experimental settings. Compared with the large-scale HT1 target library, these supplementary datasets provide additional activity

measurements from cellular AsCas12a editing experiments and were used to complement the primary training data. These supplementary datasets were included in the training data for DeepCas12a and, when retraining was required, for the baseline models under the same data-partitioning strategy. For binary classification, labels were assigned based on background-corrected indel frequencies: samples exhibiting a >2-fold change over controls and an absolute indel frequency >50% were labeled positive (1); all others were labeled negative (0). This threshold aligns with high-efficacy classification standards used in prior CRISPR studies.[14]

To model epigenetic effects, DNA methylation (RRBS) and chromatin accessibility (DNase-seq) data for HEK293T cells were obtained from the ENCODE project[15]. Target sequences (34 bp) were aligned to the human reference genome (hg38) using Bowtie 2[16]. We identified target sites overlapping with RRBS-covered regions or DNase narrow peaks to generate binary epigenetic feature maps (presence=1, absence=0)[14]. Input sequences were encoded as 6-channel tensors. Channels 1-4 represented nucleotide identity via one-hot encoding (A, C, G, T). Channel 5 encoded DNA methylation status, and Channel 6 encoded chromatin accessibility, as shown in Figure 2.

This representation captures both local sequence context and the epigenetic landscape, reflecting the various factors which influence AsCas12a activity.

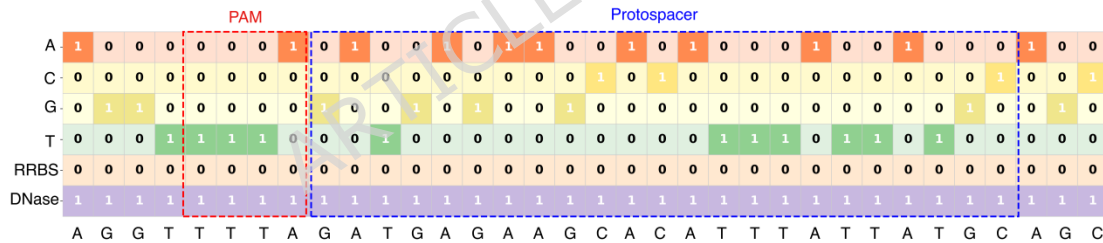


Figure 2. One-hot Encoding and Epigenetic Channel Representation of the 6x34 Input Tensor Structure. Channels 1-4 represent nucleotide identity (A/T/C/G), channel 5 indicates DNA methylation (RRBS), and channel 6 indicates chromatin accessibility (DNase-seq).

2.2 Model Architecture

DeepCas12a utilizes a hybrid neural network architecture to predict CRISPR-AsCas12a on-target cleavage efficiency. The framework comprises three primary modules: a Convolutional Neural Network (CNN) for local feature extraction, a Vision Transformer encoder (ViT) for global context modeling, and a Multi-Layer Perceptron (MLP) classifier.

Input tensors (6×34) are processed by the CNN module to extract local motif definitions and epigenetic patterns.

Convolutional feature maps are tokenized into embeddings, augmented with positional encodings, and input into the vision transformer module. A multi-head self-attention mechanism models long-range dependencies, capturing global regulatory interactions within the sequence. This design adapts the vision transformer strategy [17] to genomic data, leveraging the efficacy of attention mechanisms in sequence modeling demonstrated by architectures such as DNABERT [18].

Finally, the global class token ([CLS]) is fed into the MLP to output a probability score representing binary cleavage efficiency.

2.3 Experimental Setup and Training Parameters

DeepCas12a was implemented in Python using the PyTorch framework. Training was accelerated via CUDA 12.4 on a workstation equipped with dual NVIDIA GeForce RTX 3090 GPUs (24 GB VRAM).

Hyperparameters were systematically optimized to maximize validation performance. We employed Bayesian optimization using the Optuna library, specifically implementing the Tree-structured Parzen Estimator (TPE) algorithm. This approach was used to tune the parameters within the following predefined search spaces: learning rate (log-uniform distribution, $[1e-5, 1e-3]$), attention and global dropout rates (uniform distribution, $[0.1, 0.5]$), network depth (integer, $[4, 12]$), embedding dimension (discrete set, $\{64, 128, 256, 512\}$), MLP expansion ratio (uniform distribution, $[1.0, 4.0]$), and the number of attention heads (discrete set, $\{4, 8, 12, 16\}$). The final optimal hyperparameter configuration is detailed in Table 1. To mitigate overfitting, we applied early stopping—monitoring validation accuracy with a patience of 10 epochs and a minimum delta of 0.0—and assessed model robustness via 9-fold cross-validation on the training data.

Table 1 Selected model hyperparameters:

Hyperparameter	Value
attention_drop	0.271
dropout	0.307
depth	10
embed_dim	256
learning_rate	0.00027
mlp_ratio	2.45
num_heads	12

batch_size	256
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DeepCas12a was benchmarked against four representative AsCas12a efficiency predictors, as shown in Table 2. To ensure a fair comparison, all baseline models were evaluated using the same dataset partitions as DeepCas12a. No samples from the holdout or independent test sets were used during model training, hyperparameter optimization, or model selection.

The baseline models included:

- (1) DeepCpf1[11]: A CNN-based end-to-end predictor utilizing sgRNA sequences;
- (2) CRISPR-DT[19]: An SVM classifier integrating sequence features with manually engineered biological characteristics;
- (3) C-SVR[20]: A hybrid architecture combining CNN-based feature extraction with Support Vector Regression (SVR);
- (4) Cas12a_predictor[21]: A Random Forest model incorporating sgRNA sequences and biological features.

Table 2 Summary of mainstream AsCas12a efficiency prediction models, their references and implementation sources. Model results are obtained from either the original publications or official code repositories using the same dataset and evaluation pipeline for consistency.

Model Name	Reference	Method Type	Code Availability
DeepCpf1	Kim et al., 2018[11]	CNN	GitHub: https://github.com/MyungjaeSong/Package Library
CRISPR-DT	Zhu et al., 2019[19]	Feature Engineering + SVM	Paper Reproduction
C-SVR	Ameen et al., 2021[20]	CNN + SVR	Paper Reproduction
Cas12a_predictor	O'Brien et al., 2023[21]	Feature Engineering + RandomForest	GitHub: https://github.com/gburgio/Cas12a_predictor

3. Results

3.1 Model performance on holdout test set

We evaluated DeepCas12a on the independent test set described in Section 2.1. The model achieved an Average Precision (AP) of 0.783 and an Area Under the ROC Curve (AUC) of 0.868.

To assess quantitative predictive accuracy, we computed Spearman (ρ) and Pearson (r) correlation coefficients between predicted probabilities and experimental cleavage scores, yielding 0.630 and 0.631, respectively. These correlations indicate that DeepCas12a captures both the rank order and magnitude of cleavage efficiency, supporting its utility for quantitative prediction.

3.2 Comparison with current sgRNA efficiency prediction models

To evaluate DeepCas12a, we benchmarked it against four state-of-the-art Cas12a efficiency predictors: DeepCpf1, CRISPR-DT, C-SVR, and Cas12a_predictor. DeepCpf1 and C-SVR rely on sequence-derived deep learning features, whereas CRISPR-DT and Cas12a_predictor utilize handcrafted biological features within traditional machine learning frameworks. Baseline models were evaluated on the independent test set using the authors' public implementations and default hyperparameters to ensure fair comparison.

In binary classification tasks, DeepCas12a achieved the highest performance metrics (AP = 0.783, AUC = 0.868; Figure 3c-d). It outperformed the second-best model, DeepCpf1 (AP = 0.755, AUC = 0.858), while other baselines showed significantly lower discrimination.

To assess quantitative ranking accuracy, we computed Spearman rank correlation(ρ) and Pearson linear correlations(r) between predicted and experimental efficiencies. DeepCas12a yielded the highest correlation coefficients ($\rho=0.630$, $r=0.631$). These values exceed those of DeepCpf1 ($\rho=0.612$, $r=0.594$) and other baselines (Figure 3a-b), indicating a more accurate global ordering of target efficacy. The robustness of these comparisons across five independent runs with different random seeds is summarized in Supplementary Figure 1 and Supplementary Table 1.

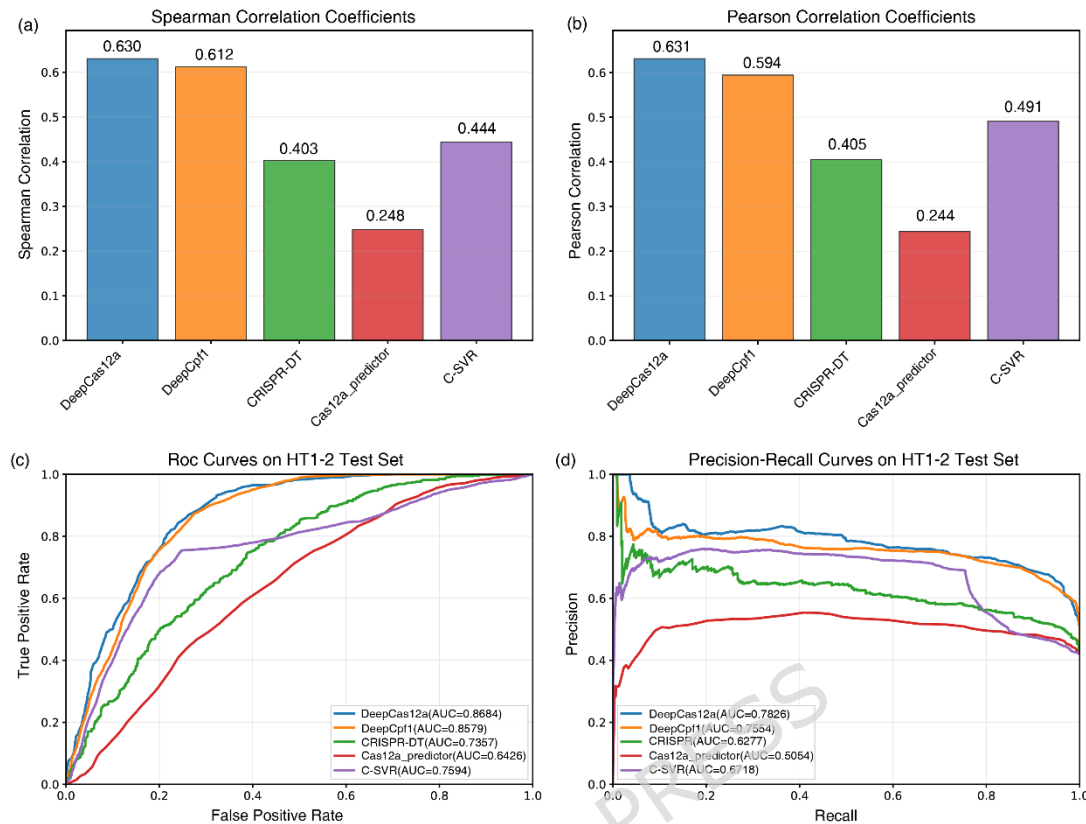


Figure 3. Benchmarking DeepCas12a against Existing gRNA Efficiency Prediction Models. (a) Spearman correlation coefficients and (b) Pearson correlation coefficients between predicted and observed cleavage efficiency (c) Precision-recall and (d) ROC curves comparing DeepCas12a with DeepCpf1, C-SVR, CRISPR-DT, and Cas12a_predictor. DeepCas12a achieves the highest AP and AUC values. The results from five independent runs with different random seeds are summarized in Supplementary Figure 1 and Supplementary Table 1.

In summary, DeepCas12a demonstrates consistent improvements over existing state-of-the-art methods in both classification and regression tasks.

3.3 Generalization evaluation on independent test sets

To assess generalization beyond the primary test set, we evaluated DeepCas12a on extra two independent datasets, HT2 (n=2,963) and HT3 (n=1,251), which were excluded from model training. These datasets were also derived from the Kim et al. (2018) study[11]

DeepCas12a demonstrated superior correlation and classification performance across both datasets compared to baseline models (Figure 4). Specifically, it achieved the highest rank correlations with experimental

cleavage efficiencies: Spearman coefficients (ρ) were 0.538 (HT2) and 0.313 (HT3), while Pearson coefficients (r) were 0.540 (HT2) and 0.294 (HT3) (Figure 4a-b). These results suggest that the model generalizes effectively to unseen data.

In classification tasks, DeepCas12a consistently showed the highest discrimination (Figure 4c-f). On HT2, it achieved an AUC of 0.823 and Average Precision (AP) of 0.697, marginally surpassing DeepCpf1 (AUC = 0.819) and significantly outperforming C-SVR (AUC = 0.751). A similar trend was observed on HT3, where DeepCas12a maintained leading performance (AUC = 0.726, AP = 0.360). These results demonstrate the model's robustness to dataset shifts.

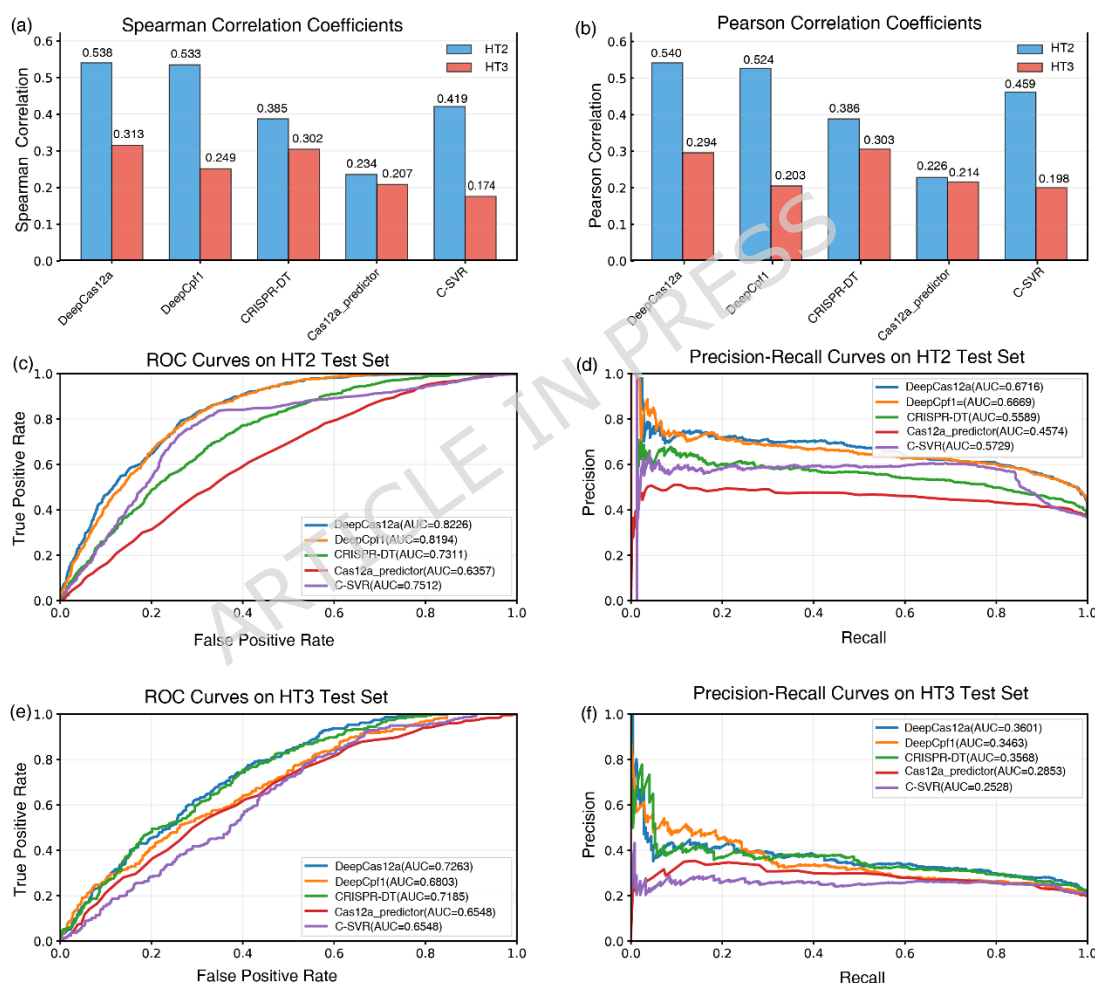


Figure 4. Generalization Performance of DeepCas12a on Independent Test Datasets. (a-b) Spearman and Pearson correlation coefficients on HT2 and HT3 datasets, showing that DeepCas12a achieves the highest consistency with experimental data. **(c-f)** ROC and precision-recall curves on HT2 and HT3 test sets, where DeepCas12a outperforms

existing models, demonstrating superior generalization ability. The results from five independent runs with different random seeds are summarized in Supplementary Figure 1 and Supplementary Table 1.

3.4 Ablation study

We performed ablation studies on the independent test set to quantify the contributions of specific feature types and architectural components. Five variants were evaluated: (1) DeepCas12a (Full); (2) Seq-Only (excluding epigenetic features); (3) CNN-Only (excluding the Transformer); (4) 50 bp context; and (5) 20 bp guide-only.

The full model achieved the highest correlation with experimental cleavage efficiencies ($\rho=0.630$, $r=0.631$; Figure 5a-b). Removing epigenetic features resulted in a minor performance decline ($\rho=0.621$, $r=0.621$), suggesting that chromatin accessibility and methylation data provide limited but complementary information.

Regarding input length, the 34 bp window (4 bp upstream, PAM, 23 bp protospacer, 3 bp downstream) proved optimal, outperforming 50 bp and 20 bp configurations. Extending the window to 50 bp likely introduced noise, while restricting it to the 20 bp guide sequence failed to capture essential flanking signals.

DeepCas12a also led in classification metrics (AUC = 0.868, AP = 0.783; Figure 5c-d). Removal of the Transformer encoder caused a distinct decrease in performance, supporting the utility of modeling long-range dependencies. These results indicate that sequence context, epigenetic features, and the hybrid CNN-Transformer architecture collectively contribute to predictive accuracy.

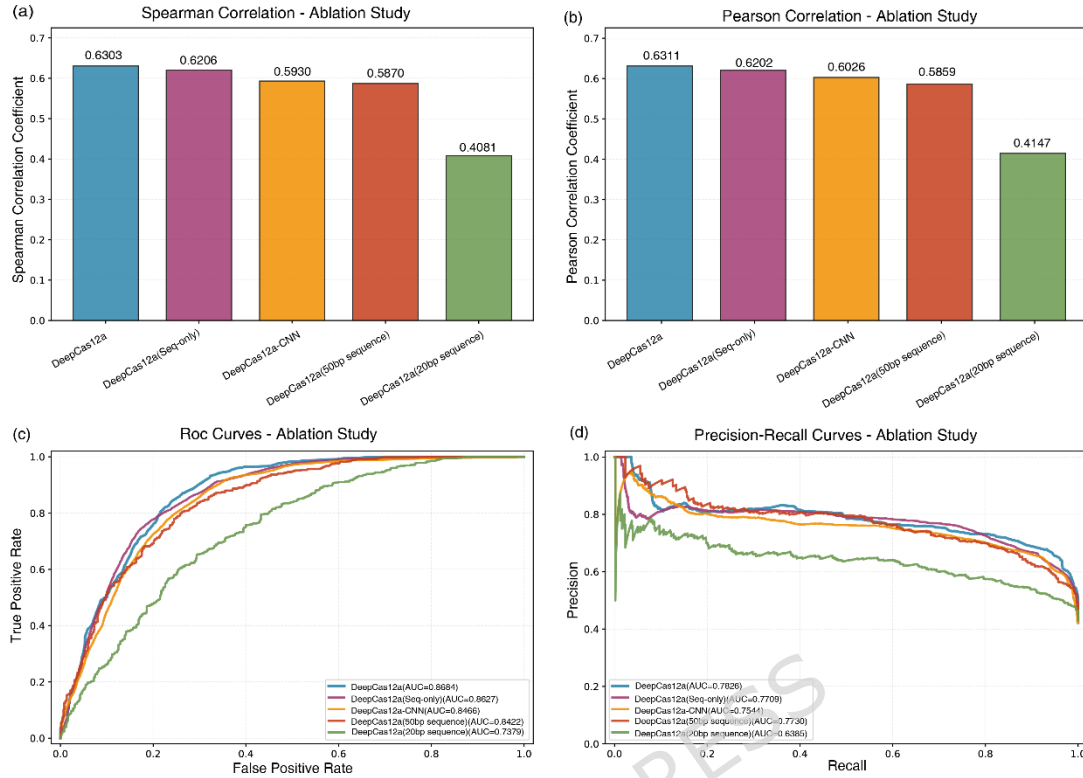


Figure 5. Ablation Study of DeepCas12a on the HT1-2 Test Set. (a-b) Spearman and Pearson correlation coefficients of different model variants, showing that the full DeepCas12a integrating both sequence and epigenetic features achieves the best correlation with experimental data. **(c-d)** ROC and precision-recall curves of ablation variants, indicating that removing epigenetic information, shortening/lengthening the input window, or excluding the transformer encoder reduces predictive performance.

3.5 Interpretability analysis via saliency maps

We visualized multi-feature saliency heatmaps for representative high-efficiency (positive) and low-efficiency (negative) targets to elucidate feature contributions (Figure 6a-b). Red regions indicate positive attribution (promoting cleavage), while blue regions indicate negative attribution (inhibiting cleavage).

(1) PAM Recognition: In high-efficiency targets (Figure 6a), the PAM core (positions 5-7) exhibits contiguous positive attribution for Thymine, whereas low-efficiency targets (Figure 6b) display inhibitory signals at these sites. This confirms the model successfully learned the preference for a T-rich PAM. At position 8, the model assigns positive weights to non-T bases

(A/C/G) and negative weights to T, reflecting the biological requirement for a V (A/C/G) nucleotide at the terminal PAM position[10].

(2) Seed Region Sensitivity: High-efficiency targets display concentrated positive saliency within the protospacer "seed" region (positions 9–14). This aligns with experimentally validated seed regions where mismatches drastically reduce AsCas12a activity[22, 23]. Conversely, low-efficiency targets show dispersed, weak saliency, indicative of ineffective guide–target pairing. Attribution magnitudes diminish beyond position 25, consistent with the minimal impact of distal nucleotides on cleavage.

(3) Epigenetic Modulation: The DNase accessibility channel predominantly shows positive attribution (red) in high-efficiency targets—particularly PAM-proximal regions—suggesting open chromatin enhances accessibility. In contrast, the DNA methylation (RRBS) channel appears predominantly negative (blue), capturing the repressive effect of local methylation.

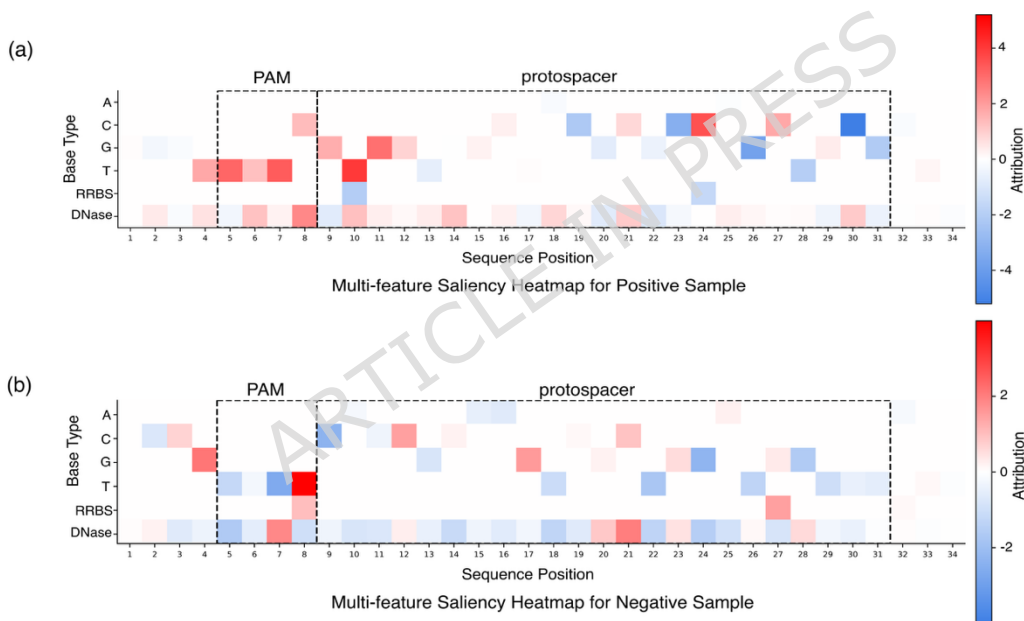


Figure. 6 Interpretable Modeling of Sequence and Epigenetic Features in Predictions. (a) Multi-feature saliency heatmap for positive sample. (b) Multi-feature saliency heatmap for negative sample.

These patterns indicate that DeepCas12a has learned biologically relevant rules, including PAM specificity, seed region sensitivity, distal tolerance, and the inhibitory influence of DNA methylation.

4. Discussion

DeepCas12a demonstrates improved accuracy in predicting CRISPR-AsCas12a on-target cleavage efficiency. It outperforms existing baselines (DeepCpf1, CRISPR-DT, C-SVR, and Cas12a_predictor) in both classification metrics (AP and AUC) and quantitative prediction metrics, including Spearman rank correlation (ρ) and Pearson linear correlation (r). The integration of local convolutional features with global attention mechanisms enables robust generalization by capturing multi-level regulatory signals. By incorporating epigenetic features, including DNA methylation and chromatin accessibility, the architecture maintains stability where sequence-only models may falter, highlighting the value of multimodal integration in AsCas12a on-target efficiency prediction.

Furthermore, it is essential to highlight the specific utility of the newly introduced ViT encoder within the DeepCas12a framework. Traditional CNN models (such as DeepCpf1) are constrained by fixed-length convolutional kernels and primarily excel at extracting local motif features, particularly around the PAM region. However, AsCas12a cleavage activity is not governed solely by local sequences; rather, it is globally regulated by the complex interactions across the crRNA-target DNA duplex. To address this, DeepCas12a incorporates the multi-head self-attention mechanism of the ViT architecture. This mechanism transcends spatial distance limitations, enabling global context modeling of the entire 34-bp target sequence. Consequently, it accurately captures the non-linear impacts of long-range dependencies on cleavage efficiency. This theoretical advantage is empirically validated by our ablation studies (Section 3.4), which show that removing the ViT encoder results in a significant decline in predictive performance, especially in the Spearman correlation. These findings demonstrate that the ViT encoder can effectively integrate local features into higher-order biological representations, endowing the model with superior generalizability and predictive robustness compared to pure CNN models when processing highly diverse sequence data.

Additionally, saliency analysis confirms that DeepCas12a independently identifies critical functional regions, including the PAM (positions 5–8) and the seed region (positions 9–14). Gradient distributions align with established biological constraints, such as the requirement for a terminal "V" in the TTTV PAM and mismatch tolerance in distal regions. Practically, researchers can prioritize targets with high predicted efficacy and favorable saliency profiles to improve experimental validation rates. This approach is particularly

valuable for therapeutic design, where identifying high-activity sites minimizes resource expenditure. Additionally, this "interpretation-validation" loop facilitates hypothesis generation—such as verifying causal links between PAM-proximal features and activity—accelerating precision functional genomics.

Despite these advances, limitations persist. First, complex cellular contexts—including cell-type specific chromatin states and transcription factor binding—may limit generalization to unseen tissues. Second, while the hybrid architecture improves representation, reliance on static high-throughput datasets risks overfitting to library-specific biases. Future work should incorporate diverse datasets from primary cells and clinical samples to evaluate cross-condition adaptability. Moreover, current reliance on bulk ENCODE data lacks temporal resolution; integrating single-cell multi-omics (e.g., scATAC-seq) could refine epigenetic modeling.

The Vision transformer architecture also positions the framework for future generative design and causal inference tasks. Generative extensions (e.g., diffusion models) could enable the de novo design of optimal guide RNAs under functional constraints[24]. Furthermore, attention mechanisms offer pathways to uncover regulatory factors, facilitating a shift from correlation discovery to causal mechanism elucidation[25, 26]. In addition, the combination of pre-training and self-supervised learning strategies is expected to enhance the model's generalization ability on rare sequences or novel enzyme variants, promoting a positive feedback loop between high-throughput virtual screening and experimental validation and accelerating innovation in CRISPR applications.

In summary, DeepCas12a provides an accurate, interpretable solution for AsCas12a efficiency prediction, establishing a foundation for intelligent gene editing design and mechanistic research.

5. Conclusion

In this study, we present DeepCas12a, a multimodal deep learning framework integrating sequence and epigenetic contexts to predict CRISPR-AsCas12a on-target cleavage efficiency. By synergizing local convolutional feature extraction with global attention mechanisms, the model achieves state-of-the-art performance, robustness, and biological interpretability.

DeepCas12a offers an effective tool for high-throughput target screening and rational guide RNA design, facilitating applications in precision gene editing and therapeutic development. Future integration of diverse multi-omics datasets and generative modeling strategies will further expand the utility of

deep learning in CRISPR functional genomics.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

Data availability

The datasets analyzed during the current study are available on the Github at <https://github.com/bm2-lab-submission/DeepCas12a>.

Code availability

The model and model usage are available on the Github at <https://github.com/bm2-lab-submission/DeepCas12a>.

Competing interests

The authors declare no competing interests.

Funding

This study was supported by National Natural Science Foundation of China [62002265]; Tongji University "Medicine + X" Cross Research Program [2025080107].

Authors' contributions

Y.S., J.K.Y., S.N., J.L.Y., D.Y. and G.C. conceived and designed the study. Y.S.

designed and trained the model. S.N. implemented the benchmarking. Y.S. and J.L.Y. analyzed the results and generated the figures. Y.S., J.K.Y., D.Y. and G.C. wrote the manuscript. G.C. and D.Y. supervised the entire project. All authors read and approved the final manuscript.

Acknowledgements

We are grateful to all the investigators and participants of this study. We also acknowledge the computational resources and support provided by Department of Infectious Dermatoses, Center of Infectious Skin Diseases, Shanghai Skin Disease Hospital.

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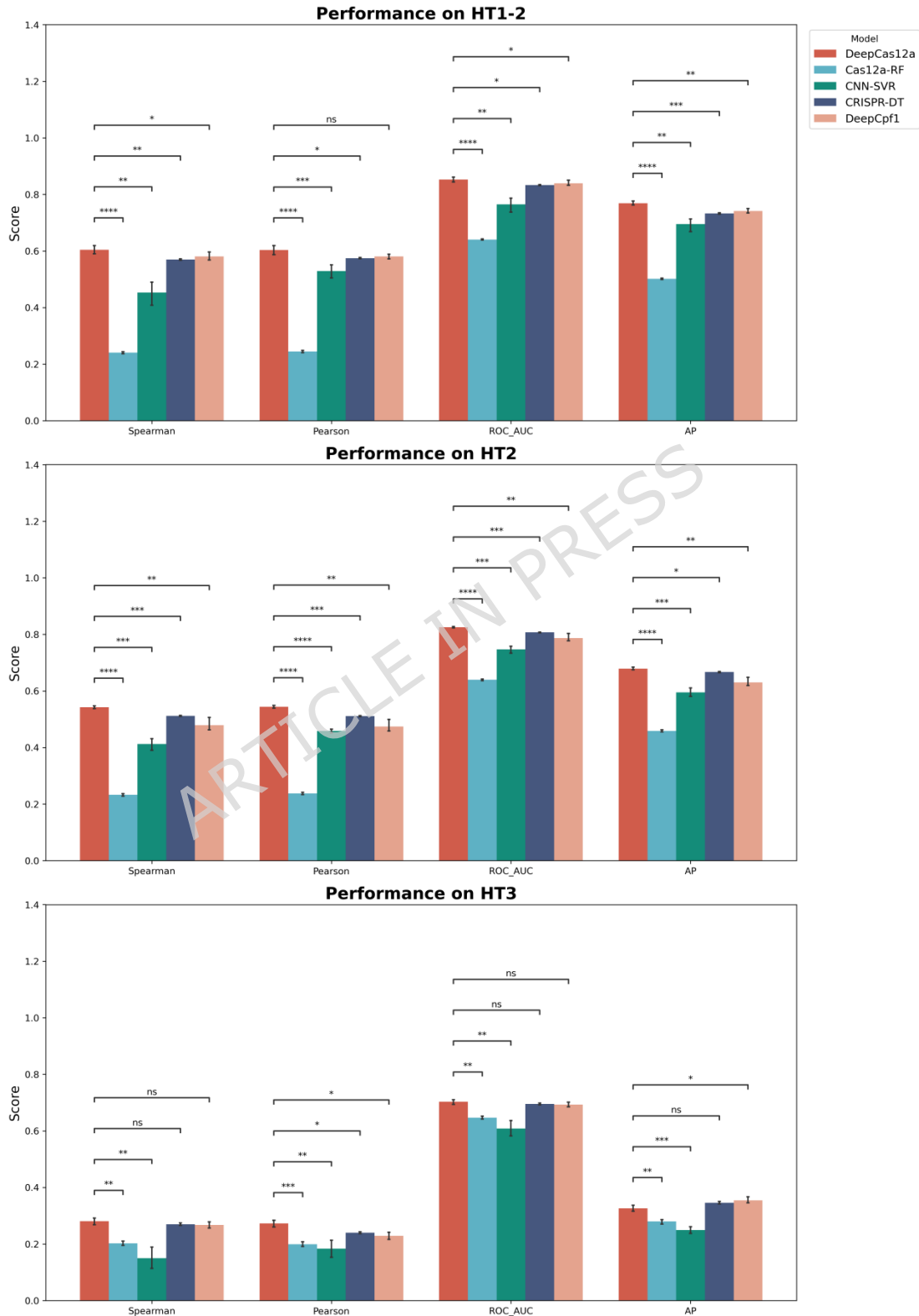
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Supplementary Figure 1. Performance comparison of DeepCas12a and competing models on HT1-2, HT2, and HT3 with statistical significance analysis

Supplementary Table1. Detailed benchmarking results of DeepCas12a and baseline models on HT1-2, HT2, and HT3 datasets. Values are presented as mean $\pm$ standard deviation. P values were calculated using paired t-tests comparing DeepCas12a with each baseline model under the same evaluation setting.

Dataset	Metric	DeepCas12a (mean $\pm$ SD)	Baseline model	Baseline (mean $\pm$ SD)	Paired t-test p value vs DeepCas12a
HT1-2	Spearman	0.6045 $\pm$ 0.0175	Cas12a-RF	0.2409 $\pm$ 0.0047	2.89×10^{-6}
HT1-2	Spearman	0.6045 $\pm$ 0.0175	CNN-SVR	0.4531 $\pm$ 0.0547	1.36×10^{-3}
HT1-2	Spearman	0.6045 $\pm$ 0.0175	CRISPR-DT	0.5700 $\pm$ 0.0028	9.19×10^{-3}
HT1-2	Spearman	0.6045 $\pm$ 0.0175	DeepCpf1	0.5809 $\pm$ 0.0191	2.79×10^{-2}
HT1-2	Pearson	0.6036 $\pm$ 0.0199	Cas12a-RF	0.2443 $\pm$ 0.0051	4.64×10^{-6}
HT1-2	Pearson	0.6036 $\pm$ 0.0199	CNN-SVR	0.5294 $\pm$ 0.0298	1.97×10^{-4}
HT1-2	Pearson	0.6036 $\pm$ 0.0199	CRISPR-DT	0.5748 $\pm$ 0.0024	2.57×10^{-2}
HT1-2	Pearson	0.6036 $\pm$ 0.0199	DeepCpf1	0.5806 $\pm$ 0.0108	5.20×10^{-2}
HT1-2	ROC_AUC	0.8526 $\pm$ 0.0106	Cas12a-RF	0.6408 $\pm$ 0.0027	3.24×10^{-6}
HT1-2	ROC_AUC	0.8526 $\pm$ 0.0106	CNN-SVR	0.7648 $\pm$ 0.0319	1.40×10^{-3}
HT1-2	ROC_AUC	0.8526 $\pm$ 0.0106	CRISPR-DT	0.8332 $\pm$ 0.0016	1.18×10^{-2}
HT1-2	ROC_AUC	0.8526 $\pm$ 0.0106	DeepCpf1	0.8395 $\pm$ 0.0112	3.23×10^{-2}
HT1-2	AP	0.7692 $\pm$ 0.0092	Cas12a-RF	0.5014 $\pm$ 0.0032	5.11×10^{-7}
HT1-2	AP	0.7692 $\pm$ 0.0092	CNN-SVR	0.6950 $\pm$ 0.0275	2.84×10^{-3}

HT1-2	AP	0.7692 ± 0.0092	CRISPR-DT	0.7331 ± 0.0029	9.46×10^{-4}
HT1-2	AP	0.7692 ± 0.0092	DeepCpf1	0.7417 ± 0.0102	5.09×10^{-3}
HT2	Spearman	0.5425 ± 0.0056	Cas12a-RF	0.2327 ± 0.0058	1.68×10^{-7}
HT2	Spearman	0.5425 ± 0.0056	CNN-SVR	0.4125 ± 0.0267	1.76×10^{-4}
HT2	Spearman	0.5425 ± 0.0056	CRISPR-DT	0.5120 ± 0.0023	1.01×10^{-4}
HT2	Spearman	0.5425 ± 0.0056	DeepCpf1	0.4791 ± 0.0304	8.50×10^{-3}
HT2	Pearson	0.5442 ± 0.0063	Cas12a-RF	0.2379 ± 0.0052	2.48×10^{-7}
HT2	Pearson	0.5442 ± 0.0063	CNN-SVR	0.4583 ± 0.0090	5.77×10^{-7}
HT2	Pearson	0.5442 ± 0.0063	CRISPR-DT	0.5114 ± 0.0023	1.80×10^{-4}
HT2	Pearson	0.5442 ± 0.0063	DeepCpf1	0.4741 ± 0.0285	4.04×10^{-3}
HT2	ROC_AUC	0.8252 ± 0.0034	Cas12a-RF	0.6394 ± 0.0035	1.67×10^{-7}
HT2	ROC_AUC	0.8252 ± 0.0034	CNN-SVR	0.7473 ± 0.0160	1.76×10^{-4}
HT2	ROC_AUC	0.8252 ± 0.0034	CRISPR-DT	0.8069 ± 0.0014	1.01×10^{-4}
HT2	ROC_AUC	0.8252 ± 0.0034	DeepCpf1	0.7872 ± 0.0183	8.50×10^{-3}
HT2	AP	0.6791 ± 0.0062	Cas12a-RF	0.4588 ± 0.0048	2.60×10^{-8}
HT2	AP	0.6791 ± 0.0062	CNN-SVR	0.5950 ± 0.0183	9.53×10^{-4}
HT2	AP	0.6791 ± 0.0062	CRISPR-DT	0.6666 ± 0.0025	2.70×10^{-2}
HT2	AP	0.6791 ± 0.0062	DeepCpf1	0.6298 ± 0.0209	7.06×10^{-3}
HT3	Spearman	0.2810 ± 0.0160	Cas12a-RF	0.2026 ± 0.0092	1.58×10^{-3}
HT3	Spearman	0.2810 ± 0.0160	CNN-SVR	0.1495 ± 0.0480	1.19×10^{-3}

HT3	Spearman	0.2810 ± 0.0160	CRISPR-DT	0.2698 ± 0.0061	2.83×10^{-1}
HT3	Spearman	0.2810 ± 0.0160	DeepCpf1	0.2671 ± 0.0131	2.47×10^{-1}
HT3	Pearson	0.2732 ± 0.0142	Cas12a-RF	0.1995 ± 0.0101	9.95×10^{-4}
HT3	Pearson	0.2732 ± 0.0142	CNN-SVR	0.1832 ± 0.0370	2.22×10^{-3}
HT3	Pearson	0.2732 ± 0.0142	CRISPR-DT	0.2399 ± 0.0041	1.39×10^{-2}
HT3	Pearson	0.2732 ± 0.0142	DeepCpf1	0.2293 ± 0.0172	2.24×10^{-2}
HT3	ROC_AUC	0.7032 ± 0.0116	Cas12a-RF	0.6464 ± 0.0066	1.58×10^{-3}
HT3	ROC_AUC	0.7032 ± 0.0116	CNN-SVR	0.6081 ± 0.0347	1.19×10^{-3}
HT3	ROC_AUC	0.7032 ± 0.0116	CRISPR-DT	0.6950 ± 0.0044	2.83×10^{-1}
HT3	ROC_AUC	0.7032 ± 0.0116	DeepCpf1	0.6931 ± 0.0095	2.47×10^{-1}
HT3	AP	0.3265 ± 0.0134	Cas12a-RF	0.2797 ± 0.0096	4.29×10^{-3}
HT3	AP	0.3265 ± 0.0134	CNN-SVR	0.2493 ± 0.0152	3.12×10^{-4}
HT3	AP	0.3265 ± 0.0134	CRISPR-DT	0.3461 ± 0.0058	5.18×10^{-2}
HT3	AP	0.3265 ± 0.0134	DeepCpf1	0.3545 ± 0.0135	3.17×10^{-2}